

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

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DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
December 2025	Process and Objective, Control Assessment Questions	Planning - Process Objective, Key Control Question - SD001.P, SD003.P, SD005.P, SE001.P, and SE008.P	Jay Mendoza	Process Objective, Control Assessment Questions SD001.P, SD003.P, SD005.P, SE001.P, and SE008.P- replaced Software Services and Software Group Services to Technology Expert Labs (TEL) Services. Software Services and Software Group Services are now referred to as TEL.
	Process and Objective	Process Objective	Jay Mendoza	Process Objective - included publisher page reference for more information on TEL Services and reference to education materials for IBM Cloud.
	Process and Objective	Planning: Data Analysis	Jay Mendoza	Planning: Data Analysis - reworded "Risk Evaluations" with "Issue Evaluations".
	Process and Objective	Planning: Data Analysis	Jay Mendoza	Planning: Data Analysis - reworded "MSAC review" with "Quarterly Close Review". Edited anomalies to provide a more concise objective of the review of the Quarterly Close Review, i.e. that some issues may indicate risks not yet addressed by management. Updated instructions to replace WWBCIT self-serve document to Open Pages Self-Serve document.
	Process and Objective	Planning: Data Analysis	Jay Mendoza	Planning: Data Analysis - reworded claims process with Concessions process
	Process and Objective	Planning: Data Analysis	Jay Mendoza	Planning: Data Analysis - added under Revenue Recognition a suggestion to include an analysis of negative GP contracts to determine any potential delivery issues
	Process and Objective	Planning: Self-Service Reports	Jay Mendoza	Planning: Self-Service Reports - replaced MSAC and Risk Evaluations with Quarterly Controls Review and Issue Evaluations respectively.
	Control Assessment Questions	Key Control Question: Contractual Agreement - SD003.P	Jay Mendoza	Control Assessment Questions - Contractual Agreement - SD003.P - suggested a test to check if there are mechanisms in ensuring that sensitive contracts are disclosed to authorized parties only.
	Control Assessment Questions	Key Control Question: SW Evaluations - SE002.P	Jay Mendoza	Control Assessment Questions - SW Evaluations - SE002.P - added a note that the process requires customer confirmation to remove or destroy software if the decision of the customer is such after SW evaluation.
	Control Assessment Questions	Key Control Question: SOD - SE005.S	Jay Mendoza	Control Assessment Questions - SOD - SE005.S - added a note that secondary controls should be documented in the SOD Assessment when conflicting tasks are identified.
September 2025	Planning	Audit Program Risk Alignment	CA&AS Ops	Changes to audit risk levels from recent revalidation

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November 2024	Process and Objective	Processes and Objective	Jay Mendoza	Updated the reference to non-channel deals audit program (Ecosystem Management) for clarity. Removed reference to Opportunity Management as details of this are added in the Planning: Read Me First Section.
	Planning	Data Analysis		Replaced SOX Disclosures with Known Issues
	Planning	Reference Material		Added a reference on the EMEA processes which have contents applicable to other geographies. Added clarity on Product Offerings and how to obtain the latest division code mapping.
	Planning	Read Me First		Described the delineation between Opportunity Management and CRM for HW and SW. Added clarity on exclusion of TLS, coverage on OIO and Enterprise License Agreements (ELA) contracts.
	Control Assessment Questions	SE001.P – Solution Delivery Implementation		Added a suggested validation of reversal to revenue accruals to determine potential accelerated revenue.

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DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
November 2023	Planning: Reference Material	Reference Material	Jay Mendoza	Updated the reference link for CRM information materials. The previous link was outdated.
	Planning: Reference Material	Read Me First	Jay Mendoza	Updated to replace any reference to customer fulfillment, O2C, and O2O to Quote to Cash (Q2C)
	Control Assessment Questions - Sales Management	SD001.P	Jay Mendoza	Updated to provide the full definition of CTEBA
	Control Assessment Questions - Sales Management	SD002.P	Jay Mendoza	Updated to replace any reference to customer fulfillment, O2C, and O2O to Quote to Cash (Q2C)
	Control Assessment Questions - Sales Management - OIO	SD004.P	Jay Mendoza	Replaced the brand names to reflect the current nomenclature of brands i.e., Infrastructure, Software, IBM Services, and IGF
	Control Assessment Question - Solution Delivery	SE003.P	Jay Mendoza	Added the full name of ILMT - IBM License Management Tool
	Control Assessment Questions - Sales Management and Solution Delivery	SD003.P, SE001.P, SE004.P, SE008.P	Jay Mendoza	Updated the linkage and reference to CRMfS audit programs. Previous references were outdated.
	SWG Services Specific	SE008.P	Jay Mendoza	Added additional test on labor claiming accuracy in relation to mapping information with expense claim information
	Process and Objective	Process and Objective	Jay Mendoza	Added Digital Sales in the introductory paragraph
	Planning	Reference Material	Jay Mendoza	Updated the reference link for CRM information materials as suggested by Elena
	Planning	Fraud Template	Jay Mendoza	Added Unauthorized Business Commitments to the first fraud risk in the table
	Control Assessment Question - Solution Delivery	SE004.P	Jay Mendoza	Added "Concessions may also be a good indicator for any solution design issues or concerns."

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DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
	Control Assessment Question - Solution Delivery	SE002.P	Jay Mendoza	For software evaluations longer than 150 days, edited to indicate "validate that appropriate and required approvals were provided." The previous reference (Market approval) was no longer applicable.
	Control Assessment Question - Solution Delivery	SE004.P	Jay Mendoza	Indicated "subsequently as required in the process" to align with available resources of FOP.
	Planning	Reference Material - Read Me First	Jay Mendoza	Added links to instructional materials for WWBCIT data, pricing delegation, and standard contract terms that can be obtained through self-serve means.
May 2023	Control Assessment Questions	SD001.P	Jay Mendoza	Added pre-sales note at bottom
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

Customer Relationship Management (CRM) for Hardware (HW) and Software (SW) covers the range of business processes from the Opportunity to the Order (O2O) and from the Order to Cash (O2C). Specifically, the most significant business processes from a risk perspective are: sales management and solution delivery, revenue recognition, supplier management, customer claims and complaints, concessions, and financial management. The CRM process enables the design and delivery of a solution according to a customer's conditions of satisfaction, business needs, and required quality standards. After integrating all solution components, quality assuring and pricing the solution, a proposal is presented to the customer. Once the solution is accepted, the solution delivery phase starts with a review of IBM commitments. The solution is then implemented and, in parallel, managed to ensure a quality delivery of IBM commitments in a profitable manner, while ensuring customer satisfaction. Renewal or additional opportunities are also identified to extend or add services. The prime business objective of the CRM process is to develop and deliver the best value business solution meeting our customer requirements. HW, SW, and related TEL services can be sold direct or through channels such as business partner (BP) or Digital Sales. This program covers HW and SW deals that are non-channel deals (i.e. Direct deals, Digital Sales deals, etc.). For references to transactions through BPs, please refer to audit program 56-00-00 (Ecosystem Management) for both HW and SW.

Link to TEL Process Catalogue: <https://w3.ibm.com/w3publisher/software-services-operations/new-page>

Link to IBM Cloud References: <https://ibm.seismic.com/app#/doccenter/861ea1fd-99e0-44d7-9135-85412e5c28d1/doc/%25252Fdd3359e5f7-a856-a91b-7688-41024b2ac637%25252FdfNTY4NmVhOWItY2RkNS04ZWY3LTZkNzltZTQwZjczMWUyMjk1%25252CPT0%25253D%25252CUHVibGJIENsb3VklFBsYXRmb3Jt%25252FdfNDRmODBiMzMtY2ViMC0zMdI1LTVhNDEtNzg2OTg4MWVmZDBl%25252COthers%25252FdfOTRiYmU4NTQtNWY4NC03Y2QyLWZjYWUtOGlxYmFmZjkyZThk%25252CPT0%25253D%25252CUm91bmR1cCBvciBzdW1tYXJ5%25252Ff06a3368c-47b2-4735-91a1-b9c93e7423c2//?mode=view&searchId=35c23349-7e7d-47b1-afaa-26380fdc7208&anchorId=b54d964e-9a19-46db-915f-ef4afb9d184b>

Important note on the scope of the audit:

At the beginning of the audit, it should be decided whether transactions that went through Digital Sales are included in the scope of the audit.

Note: the SW organization, products, and processes are particularly dynamic and subject to transformation. Changes that might have occurred in the prior months before the engagement should be understood at an early stage of the audit and considered for definition of the process interviews calendar.

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PLANNING: Audit Program Risk Alignment

Customer Relationship Management for Hardware and Software

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Financial Management	High	Solution Delivery	SE004.P, SE006.P
Solution Delivery	Medium	Solution Delivery	SE001.P, SE002.P, SE003.P, SE005.S, SE007.P, SE008.P
Sales Management	Medium	Sales Management	All

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Overall Business Performance Metrics	Review business results and compare: YOY Revenue and GP Review customer satisfaction results	- Areas with positive and negative growth and declining GP - Customer dissatisfaction	- Overall business health - Not meeting customer expectation
Management Reporting (including operational performance measures)	Review the monthly/quarterly management information reports used to monitor key financial and operational objectives and performance during the period. Review compliance testing results.	- Management reporting may not be available - May highlight low performing contracts for review - May provide detailed information and/or root cause analysis on low performing contracts - Compliance testing may highlight missing mandates, billing issues, asset management issues etc.	- Potential process weaknesses - Key metrics may be missed - Out of line situations not addressed
Issue Evaluations (formerly risk evaluations)	Review the nature of any issue evaluations raised by the business during the audit period and/or any pending approval.	- Required approvals may not have been obtained - Committed deadlines may not have been met	Potential out of line situations and/or process weaknesses

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Quarterly Close Review (formerly Management Self-assessment of controls MSAC)	Review quarterly close review reports and understand the issues disclosed and highlighted.	Some issues may indicate risks not yet addressed.	Potential process weaknesses and out of line situations
Business Controls/Peer and Other Reviews	Using all reviews provided in the IDR, understand the findings identified and the actions implemented to address the findings.	May highlight pricing violations, work at risk, sales management and solution delivery mandates missing	Potential out of line situations
Known Issues	Review detail of known issues.	May indicate the provision of free services as claims avoidance, or bypass of the Concessions process	- Unauthorized commitments - Delivery issues - Process bypasses
Sales Management and Solution Delivery	- Credit and Denied Parties List (DPL) Checks - Technical Delivery Assessment (TDA) Pricing / Special Bids Process - Firm Order Exception (FOE) Policy - Enterprise License Agreements (ELAs) - Frame Agreements/ Global Customer Offerings (GCO) - Labor Claim	- TDAs not performed - Labor claimed prior to contract signature without an authorized Pre-Contract Work (PCW)	- Opportunities not adhered to the sales management process, or incorrect management information - Improper assessment of risk - Potential delivery deficiencies - Potential sales management issue - Potential revenue misstatement - Ineffective controls - Inappropriate labor claiming - Unbilled revenue/ unrecoverable costs

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Asset Management (Software Evaluation Offering [SEO] and SW License Compliance [SWLC])	Review software evaluation offerings. Review SW licenses compliance universe. Review SW licenses compliance suppliers' contract and invoices paid to suppliers.	- Excessive periods and/or more equipment than detailed in exhibits - Lack of required approvals - Incomplete or inaccurate inventory records - Unauthorized settlements - Inconsistent program usage - Wrong technical methodology used by the compliance partner - Lack of independent analysis to prioritize customer selection - Lack of timing service levels and penalties for delays - Incomplete suppliers' invoice acceptance methodology	- Loss of assets - Loss of revenue - Customer dissatisfaction - Asset used by ineligible user - Erroneous results leading to image damage - Prevent effective identification of non-compliant situations - Late identification of licenses over usage and late financial incomes - Undetected deviations and financial loss
Billing and Accounts Receivable	Review reconciliations between ELA SW inventory and billing. Review credit notes report, paying attention to the customer, value, date, and reason code.	- Invoices not in line with SW inventory - High value of credit notes - Recurring credit notes	- Over or under billing - Settlements made outside policy - Potential process weaknesses - Potential fraud

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Revenue Recognition	Review ledger entries report paying attention to: • High value revenue accruals • High value cost accruals • Cost deferrals • Revenue deferrals • Provisions • Revenue reversals in the month following a quarter end • High value credit notes • Negative gross profit (GP) at contract level	• Unbilled revenue • High value accruals/deferrals • Recurring accruals/deferrals • Gross vs Net fact sheets not completed where required • Negative GP may indicate delivery issues	• Inaccurate revenue recognition • Potential customer disputes • Revenue overstatement (GvN)
Customer Satisfaction and Concession Data	Review concessions, settlements, and write offs. Review customer satisfaction measurements results and analysis and action plans.	• Low number of claims may indicate settlements outside of the claims process • High value write offs • Missing customer satisfaction measurements • Proper actions not being taken following customer satisfaction survey results	• Unauthorized commitments • Settlements outside policy • Lack of visibility over customer satisfaction • Potential disputes/customer dissatisfaction not being early identified

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PLANNING: Reference Material

At the time of CRM audit planning, contact Internal Audit CRM Process focal point to seek expertise during the audit, and also to have appropriate executive support during the review.

CRM information such as Education, Processes, Tools, On-line flow chart, Roles, Business Process Library, Enterprise Process Framework, Architecture and Design is accessible at the web address <https://ibm.seismic.com/Link/Content/DCg7cFB9fBGPC892q9b4hFTdjfB3> and <https://w3.ibm.com/w3publisher/ibm-sales-cloud>
<https://w3.ibm.com/w3publisher/emea-technology-sales-management-hw-sw>

General Data Protection Regulation (GDPR)

GDPR Readiness: <https://www.ibm.com/data-responsibility/gdpr/#4>

Compliance, Audit, Risk Mitigation & Management: <https://w3.ibm.com/w3publisher/dprc>

Links to specific processes mentioned in the audit program have to be requested to the client in order to obtain updated documents.

Product Offerings that form part of CRM for HW and SW

To help determine whether an offering belongs to CRM for Hardware or CRM for Software, you may refer to the latest division code mapping. Request for the latest extraction of AIWCW.BRAND_ROLLUP_R A from the CA&AS Data Analytics Team. Updates to the division code mapping can also be found in the EPM Planning Analytics publisher page <https://w3.ibm.com/w3publisher/epm-planning-analytics/ibm-brand-consolidation/ww-consol/dimension-updates>.

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PLANNING: READ ME FIRST - Important Note when Auditing CRM

Important note when auditing CRM

Audits of CRM focus on the end-to-end process of engagement with our clients and delivery of projects, goods, or services. The contractual relationship with the client is maintained by the owning Brand/CFU (Client Facing Unit) opportunity owner, project manager, or project executive. They are supported by multiple organizations in both design and delivery. Internal audits of CRM are end-to-end in terms of looking at Quote to Cash (Q2C) processes. The Q2C organization support business units in delivery of contracts in various degrees and this support sometimes differs by country.

It is important when conducting an audit of CRM to ensure that we do not overlap into in-depth testing of Q2C processes (for which there are separate audit programs).

For example, whilst conducting an audit in CRM, we are interested in the accurate and timely processing of a transaction end-to-end. We should not be testing the supporting Q2C processes in depth, which would normally be covered by a Q2C audit. The same concept applies to procurement and accounts payable processes. It is important to bear this in mind when scoping your audits in this area, to avoid any potential overlap with deeper process audits of Q2C or procurement whilst it is still appropriate to extend testing into any area where significant risks are identified during the engagement. Also, remember that orders that are part of a multi brand deal (e.g., OIO contracts) must be in scope in a HW or SW audit.

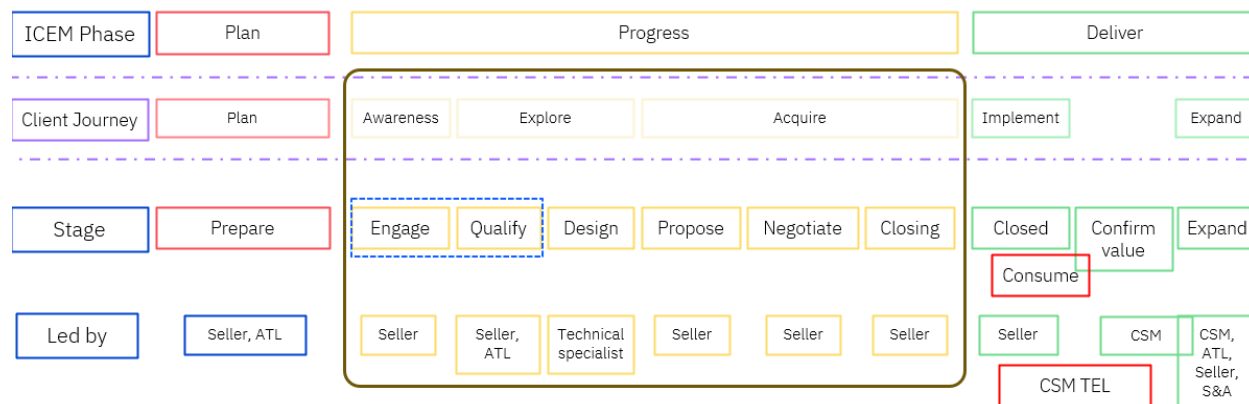
Example: Accounts Receivable

The root cause of accounts receivable exceptions (such as a large amount of disputes, aged debt or write-offs) may identify issues in sales management or solution delivery. So, the test would be to identify why the customer was not paying their invoices, not whether Accounts Receivable executed their process correctly. It is acceptable to request data from other areas as it can point to problems in the CRM process.

Opportunity Management delineation

Similarly, Opportunity Management is also covered separately in another audit program. Based on the IBM Client Engagement Model, there are three phases in every opportunity – Plan, Progress, and Deliver. Opportunity management will cover until Qualify stage. CRM will cover from Design onwards. Opportunity Management covers both field sales and digital sales (inbound sales).

Below is the IBM Client Engagement Model. Enclosed in brown box is the Opportunity to Order process.



For Q2C, refer to Audit Program 52-00-00.

For Opportunity Management, refer to Audit Program 60-10-10.

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PLANNING: READ ME FIRST - Important Note when Auditing CRM (Continued)

Additionally, Technology Lifecycle Services (TLS) belong to a separate auditable program, 60-00-00. If the unit, market, or geography being audited has a separate OIO auditable unit, the OIO transactions will be reviewed under the OIO auditable unit. Some OIO contracts may also include ELA transactions, in such case, the ELA components will also be reviewed under the OIO auditable unit. Get clarity during the audit if the universes may contain OIO transactions and obtain audit client's help to distinguish them. In certain cases, this distinction might have to be done at a transaction level.

Internal Audit are not limited by business or process during reviews, but please use this note to assist you with identifying the relevant areas of risk and extent of testing to be performed particularly in areas where there are separate auditable entities for connecting processes.

Self-Service Reports

Please contact the Internal Audit Technology team to verify what Q2C reports can be pulled independently in your geography.

Some information previously included in the Initial Data Request (IDR) items which continue to be relevant information in audits will now be extracted directly by the audit team. These are as follows:

Information previously in IDR	Instructional Materials to download information and applying for access to the source tool
- Quarterly Control Review - Issue Evaluations - Proactive Business Control Reviews - Key Control over Financial Reporting (KCFR) control points - KCFR defects	https://ibm.ent.box.com/file/1365412782200 to be updated with new reference from FBC
Pricing Delegation Repository	https://ibm.ent.box.com/file/1365421588129
Standard contract templates for CRM for HW / SW	https://www.ibm.com/support/customer/csol/terms/internal

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper, or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Side letters or Unauthorized Pricing Commitments or Unauthorized Business Commitments	Contract validation and acceptance. SOD in pricing, contract preparation, contract validation, sales	Validate if approved pricing case is in line with contract. Review credit notes and customer claims and concessions for root cause and required approvals. Review aged accounts receivable and disputed invoices for root cause and actions taken. Review contract for non-standard terms and conditions and check if required approvals were obtained. Check bid and contract documentation for correspondence with customer that could be a side commitment. Validate that all required approvals were obtained prior to proposal submission to the customer.
Contract backdating / alteration	Contract validation and acceptance	Review contractual and supporting documentation to understand dates and sequence of events and approvals. Also review for manual alterations. Review contracts signed at the end of quarter, weekends, holidays. Compare dates of approvals with dates of signature.
Fake orders	Order cancellations	Test order cancellations, verify that goods have been returned and revenue has been reversed. Investigate reason for cancellation.
Products free of charge	SW evaluation monitoring	Test if follow-up activities were performed on expired SW evaluations contracts and if there are proper business justifications and approvals for extensions. Validate timely return of products or disablement of SW for expired contracts.

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PLANNING: Fraud Template (Continued)

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Premature Revenue Recognition	Accounting Reviews	Check shipments that were for extended period in time in warehouse/distribution center and determine root cause. Review aged unbilled and overdue A/R. Revenue reversal in the subsequent month/quarter for revenue recognition. Review if information sent to Accounting for review was complete and accurate.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Objective:

Are solutions designed in accordance with customer requirements, properly reviewed and approved by management, and in compliance with IBM guidelines and instructions.

Risk/Exposure:

The design phase of a contract is crucial to the success of a project. A complete and realistic evaluation of the risks and costs associated with the offering is extremely important, and a key input is the full understanding of customers' expectations. Also, a thorough evaluation and approval of all pricing elements and non-standard terms and conditions is necessary to prevent legal issues and safeguard IBM's profitability.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: Sales Management Mandatories (SD001.P)

Objective:

To ensure that the opportunity has been progressed using the mandatory IBM processes and procedures.

Note: The sales management process starts after a decision has been made within the opportunity management process to pursue the opportunity (qualification stage).

Approach:

For both hardware and software deals - including TEL services - determine that mandatory activities required based on the complexity and/or size of the opportunity were performed. For the selected opportunities verify the following steps of the client value method (CVM):

- The pricing case is approved by delegated, authorized personnel. Where pricing approvals are required from multiple lines of business, ensure that approvals were based on a consistent understanding of the solution.
- All internal approvals were in place before proposed solution confirmed with customer
- The customer's credit worthiness has been assessed
- The customer was checked against US Export Regulation requirements
- A denied parties list check has been carried out
- Where required, legal, contract and negotiations (C&N), accounting, and other organizations have reviewed the solution, and contract as required
- Non-standard terms and conditions were properly approved
- A Technical Delivery Assessment (TDA) was performed, approved and identified risks were mitigated.
The configuration was developed using an approved method (manual design or system generated configuration) and that it was created prior to submitting the proposal/contract to the customer
- The customer requirements were clearly and comprehensively defined, and a detailed solution that addressed requirements was included within the contract and supporting statements of work
- The solution was correctly priced (refer to test SD002.P for more details). Costs were compiled for all solution components and non-standard components were included in pricing, including, e.g., trade-in, buy-back, demonstration equipment, trial-period-offering.
- Where required, the Global Accounting Checklist (GACL) has been filled and sent to accounting for review
- The contract was signed by the customer and IBM, or an authorized pre-contract work exists (for TEL Services only)

In addition, for the selected contracts verify that the customer requirements were clearly and comprehensively defined.

NOTE: During pre-sales, determine if there were engagements held at Innovation Studios (if existing in the country being audited) which were initiated by the Sales team. Engagements conducted in Innovation Studios may include product demonstrations (like Watson in Munich), proof-of-concept demonstrations, etc., that can help promote IBM's products and services to win an opportunity.

- Were the compliance items, which were owned by the Sales teams and have the responsibility, adhered to? Compliance items related to engagements held at Innovation Studios will include Compliance with Transportation, Entertainment and Business Amenities (CTEBA) approvals for attendees (where applicable), GOE identification for attendees, marketing permission were obtained per IBM guidelines (or any applicable local regulations), DPL checks, etc.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: Pricing (SD002.P)

Objective:

Determine if the solution was correctly priced and approved and if all contractual obligations of IBM were adequately reviewed and properly authorized.

Approach:

For the selected opportunities verify the following:

- The price offered to the customer was approved within the delegated limits of the pricer and the organization, including review by relevant review boards.
- All elements of pricing were approved prior to communication to the customer.
- The price was developed in accordance with published methodology and using approved pricing tools.
- The pricing case includes the correct level of revenue apportionment (relevant for cross brand deals).
- Where IBM Financing is part of the offering to the customer, are all elements of the price consistently documented, and financing based upon the final approved price?
- Late changes to the proposal price were appropriately approved and documented.
- Focus on pricing completeness for Enterprise License Agreements (ELAs) and Open Infrastructure Offerings (OIOs).
- The pricing case took into account the current contractual position, if the proposal related to an existing contract.
- There may be financial issues associated with the current contract which should be addressed in the renegotiated contract, in terms of pricing and the revised contract terms.
- Where the contract is an extension of a previous contract, check that issues identified in previous quality assurance reviews have been appropriately addressed in the pricing of the extension.
- The contract price was documented within a billing plan and transferred to Q2C accurately and in a timely manner (this relates mainly to ELA, OIO, Services and similar type of contracts).
- The pricing of the solution has been reviewed for completeness and any gaps have been identified (e.g., all business units in scope are part of costing), e.g., for OIOs.
- The complexity of the solution was identified, and the impacts on time and resources quantified.
- Where pricing includes third-party costs for services and /or OEM equipment, validate the actions taken to ensure the accuracy of those costs. Has procurement obtained commitments (in the form of proposals) from suppliers that support the pricing to IBM's customer?
- Where negotiations extend beyond the initial validity period, are proposals from suppliers revalidated? For higher complexity and large contracts, and contracts with high percentage of third-party spend, supplier quotations and terms should support both pricing and IBM customer terms and conditions (this relates mainly to ELA, OIO, Services and similar type of contracts).
- Where any open charges from a previous contract carried forward into the new contract (e.g., in case of ELA, OIO, Services and similar type of contracts).

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: Contractual Agreement (SD003.P)

Objective:

To ensure that the proposal submitted to the customer met the customer requirements and that all contractual terms and conditions have been appropriately reviewed and approved.

Determine if the designed solution matched customer requirements and IBM profit objectives. Verify that customer signed the deal before commencing the supply.

Approach:

For hardware and software deals, determine the mandatory activities required based on complexity/products/size.

Verify that they have been obtained prior to proposal issuance to the customer, and the following approvals were collected and available at that time:

- Confirm credit check/approval was still valid at time of proposal submission to client
- All conditions in the approval of the contract (e.g., by Pricing, Legal, Accounting, Quality Assurance) were met and corresponding approvals received prior offering the proposal to the customer
- Non-standard terms and conditions were explicitly approved, and all approvals were obtained before commitment to the customer.
- For contracts with non-disclosure agreement or restrictions to disclose contractual documents to defined group of individuals, ensure that there are mechanisms to comply with these terms.

For TEL Services see CRMfS, Control Assessment questions, "Contract Design and Agreement" (SD003.P).

Additional checks to verify that customer approval was obtained before commencing the supply:

- The proposal was accepted by the customer within the proposal validity dates
- Any evidence that contract signatures had been back dated or the contract had been changed by the customer before signature
- The person who signed the contract had the actual authority within the client's organization
- Verify if Purchase Order (PO) driven customers have been identified, and necessary requirements were applied. For this purpose, request copy of the PO client guidance available at the time the deal was signed.
- Were the contracts signed before commencement of supply or of service? If not, was there an approved work at risk place (in case of TEL services type of contracts) or other required exception approvals in place?

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: Open Infrastructure Offerings (OIO) (SD004.P)

Objective:

Verify that specific processes and procedures for complex cross-brand deals like open infrastructure offerings (OIO) were followed, and required approval obtained, during the sales management phase.

Approach:

For complex cross-brand deals, known as open infrastructure offerings (OIO), specific processes have been established, detailing the mandates for sales management and solution delivery. The mandates detailed in the process established at geography level are in general complementary to the usual mandates required for each line of business and managed via an oversight system. CA&AS teams testing OIO deals have to request a copy and be aware of the processes in place and local deviations for this complex, cross-brand deals, regardless the approach of reviewing them for audit purposes (e.g., a specific auditable unit has been defined or you sample OIOs during portfolio CRM audits, or both).

An OIO contract is like an umbrella pulling together several brands (Infrastructure, SW, IBM Services, and/or IBM Financing). The brands approve their brand pricing (no pricing delegation for the OIO team) that will then be incorporated into the OIO price. Each brand has to perform all required sales management steps. An OIO contract is a firm commitment from the customer to buy certain volumes of products at an agreed price. Capacity planning and pricing are performed before contract signature. OIO signings are not reported to the public. Each brand reports its own signings according to the brand signing recognition guidelines. But some OIO salespeople are paid based on OIO signings.

For the selected sample of contracts, order letters, and amendments, if required, determine that:

- Credit check was prior to customer signature
- Brand quality assurance was accurately and timely performed
- Brand prices were accurately calculated (rebates included), timely approved by the delegation holders, and consistent with the contractual terms and conditions
- Vendor proposals were in line with IBM contract
- Prices and contract details were not released to the customer before approval
- System Assurance Review Boards (SARB) reviews were performed, required participants were involved, approvals were obtained, and identified issues were assigned and tracked for resolution
- Contract and Accounting fact sheet and pricing model were available before final SARB
- The final OIO price was approved before customer signature and is consistent with the contractual obligations
- Non-standard terms and conditions (T&Cs) were reviewed by Legal and approved by appropriate functions
- All mandatory contract elements were included in the final contract
- The contract was signed by the relevant parties and after all the approvals were provided
- Contract changes (whether order letter or amendments) were appropriately approved
- Accounting was consulted to determine possible impacts on revenue recognition (brand)
- Global Revenue Allocation Tool (GRAT) analysis was performed
- Review the process documents and verify if risks in the OIO-unique processes were identified and controls were designed, monitored, and reported effectively.

Please bear in mind that orders that are part of a multi brand deal (e.g., OIO contracts) must be in scope in a HW or SW audit.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: Technology Expert Labs (TEL) Services Specific (SD005.P)

Objective:

Ensure all elements of pre-contract work were in place to protect Intellectual Property Rights (IPR) and limit liabilities. In addition, ensure payment of the work performed prior to contract signature.

Approach:

As TEL services do not constitute the core activities of the SW business, the following section will focus on pre-contract work requirements, as it was identified as the main risk potentially affecting profit of the SW business. The CRM for Services audit program includes additional information which can be used when testing sales management for TEL services deals.

For pricing, verify the following:

- The price offered to the customer was approved within the delegated limits of the pricer and the organization, including review by relevant review boards
- All elements of pricing were approved prior to communication to the customer
- The pricing includes the recommended contingencies for risks
- The pricing includes all cost considerations including costs of subcontractors, penalties, quantified risks, joint verification / inventory assessment costs
- The price was developed in accordance with published methodology and using approved pricing tools
- The complexity of support processes was included in the price. (e.g., costs associated with billing requirements, additional support from other IBM support functions, e.g., C&N, Procurement)

For Software Pre-Contract Work (PCW), verify that:

- A PCW engagement was in place
- Accurate costing on planned resources was using the official cost rates
- Documented approval from the appropriate approver prior to the commencement of any PCW was obtained – regardless of the complexity assessment of the original bid
- A one-way Preliminary Commencement Letter (PCL) or approved waiver stating that the client and IBM agree to conduct business, finalize a contract, and begin work, was sent to the client or communication from the client requesting TEL Services work prior to contract signatures was received
- A minimum set of terms for an engagement (including limitation of liability and IPR) was included in the PCL
- Any modification of the terms and conditions was reviewed with Quality Assurance and Legal
- The PCL and the engagement form were sent timely to services customer fulfillment for loading the PCW document in the contract management system (work item number)
- The cost related to contractors was considered within the existing PCW approval delegations
- Verify that work did not begin prior to the approvals or the date the commitment letter is sent to the client. Use the labor claiming universe for this purpose.
- Ensure that the signed contract covers the activities performed during the PCW period, so that those activities can be invoiced
- Verify that the work item number generated by services customer fulfillment was activated upon contract signature and a system-generated invoice was issued for the engagement
- Verify that the sales management process was completed through Bid Sign-Off before PCW was approved

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Sales Management

Primary Test: TEL Services Specific (SD005.P)

(Continued)

Review if the following approvals and verifications were completed prior a PCW request:

- Credit assessment was requested and approved under the delegations
- US Export Check was performed
- Denied Parties List check was performed
- Global Services Risk assessment or Self Risk Assessment was performed
- Opportunity owner validated all the requirements in the TEL Services Quote to Cash Process Workflow tool
- Requirements for PCW with Government Owned Entity (GOE) clients were adhered to

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Objective:

Determine if there are controls to manage the solution delivery and verify that the solution is delivered in line with the contracted solution.

Risk/Exposure:

Solution might not be delivered in line with contracted solution which could lead to financial loss and customer dissatisfaction.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Delivery Implementation (SE001.P)

Objective:

Determine if there are controls to manage the solution delivery, and to verify that the solution delivered is in line with the contracted solution.

Approach:

For the selected contracts verify the following:

- Equipment was ordered and delivered to the customer in accordance with the contract schedule and installed at the correct customer
- TEL Services were monitored against the plan and actions are taken in case of deviation
- Check to see that any orders that have been processed based on Letters of Intent from the customer or IBM initiated orders have the required authorizations
- The project activities started after the contract signature. Alternatively, a formal work at risk was requested and properly approved, pending the final contract.
- Deviations or changes to the original contract are authorized by management and agreed with the customer before execution. Review the process for making changes to the contract and determine if the change management process is adequate, effective, and timely. Verify that all changes are incorporated into the revenue, cost (including accruals), and schedule reporting system.
- Acceptance tests as required by the contract terms are performed on schedule and formally agreed by the customer
- The solution delivery phase closing process is followed and upon completion of the delivery, the resources were released, and the customer satisfaction feed-back was initiated
- Validate if there were reversals of revenue accruals that pertain to potentially inaccurate or unsupported Customer Requested Arrival Date (CRAD) or potentially inaccurate customer acceptance readiness. Unsupported CRAD referred to in this are those that are not properly documented with customer approvals or communications. This could help identify potential accelerated revenue recognition.

For general controls over TEL services delivery implementation, refer to audit program for Customer Relationship Management for Services, Control Assessment Questions, 2. Services Solution Delivery, Solution Startup and Implementation (SE001.P).

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: SW Asset Management (SW Evaluations) (SE002.P)

Objective:

Ensure that ownership of all licenses which are part of the evaluation offerings is clearly defined and the responsibility for management and control of the assets is identified. Also, verify that procedures are adequate to manage these assets and that contracts exist. Verify that controls are in place to ensure SW discontinuances at the end of the evaluation or trial period are processed in a timely manner.

Approach:

Distributed SW (DSW) evaluations (Passport Advantage), follows a worldwide process and managed centrally (local deviations might exist). Part of the process is automated, and the IBM Licenses Agreement Evaluation Program (ILAEP) Agreements are sent to the client by the SW Evaluation Program system. When the customer accepts the terms and conditions, the sales representative is notified.

For the selected DSW evaluation samples verify the following:

- The ILEAP terms and conditions were approved by the customer
- All approvals were granted timely and by the correct level of authority according to the length of the evaluation
- Extensions were managed according to the process and approved by the correct level of authority
- For software evaluations longer than 150 days the appropriate and required approvals were provided
- All approval evidence were kept in the repository
- At the end of the evaluation period, the customer decision either to terminate, extend the evaluation, or purchase the software, has been received
- In case of purchase, verify that the SAP order number has been provided by the sales representative and that the SW has been billed to the customer
- In case of no purchase, there was evidence of software removal/destruction. The customer is required to confirm, as per the DSW Evaluation process
- Verify that a local management system as well as controls are in place to avoid repeating delinquent situations related to DSW evaluation overdue, and over extended situations

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Software License Compliance (SE003.P)

Objective:

Confirm that the WW Software Licensing guidelines in effect during the audit period have been properly implemented for the closed software license reviews performed in the same audit period.

Approach:

Validate that the WW Software Licensing guidelines have been followed by reviewing required documents stored in the SWLC DB.

For the closed software license reviews, select a sample and verify that:

1. The structuring and pricing for the Software Licensing Resolutions defined in the Guidelines were followed, or approval was granted.
2. If there were exceptions, check and understand the documentation stored in the SWLC DB includes:
 - Communication to client informing them of software license review where applicable (not applicable and not limited to self-report/self-declarations, IBM License Management Tool (ILMT) reports, z analytics)
 - Report outlining client's software entitlement versus deployment position (ELP) where applicable. For self-report/self-declarations, client's written communication of excess use of software.
 - IBM resolution to settle all the non-compliant software products (net of mitigations/adjustments) identified during the compliance review
 - Required approvals obtained before resolution was issued to the client where applicable (i.e., in case of mitigations/adjustments or discounted licenses/back maintenance)
 - Close-out letter or agreement (e.g., ELA agreement) including the settlement content and terms sent to the client to formally close the review
 - Legal approval to any changes to the terms in the standard close out letter or agreement (e.g., ELA agreement)
 - Order placed by the client in line with the resolution
 - Post review follow up actions where applicable

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Revenue Recognition (SE004.P)

Note: Limit the scope of the test to what is required to get an end-to-end view of the selected contracts/orders

Objective:

To ensure that revenue recognition transactions were in accordance with generally accepted accounting principles and IBM practices.

Approach:

For TEL services, please refer to the audit program Customer Relationship Management for Services (60-00-00), Control Assessment Questions, Revenue Recognition (SF002.P).

IBM shall recognize revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which IBM expects to be entitled in exchange for those goods or services. To achieve that principle, IBM shall apply the following steps:

- Step 1: Identify the contract with a customer
- Step 2: Identify the performance obligations in the contract
- Step 3: Determine the transaction price
- Step 4: Allocate the transaction price to the performance obligations in the contract
- Step 5: Recognize revenue when or as IBM satisfies a performance obligation

For Hardware Revenue, firm order documentation consists of a written customer agreement with the following aspects of an order:

- product configuration
- customer requested arrival date (CRAD)
- delivery/install location
- price
- no contingencies (e.g., contingent upon customer's ability to secure financing, board approval, quality tests, execution of another agreement that is required to complete the arrangement like a leasing agreement, etc.)
- for customers who require purchase orders to commit and authorize funding: receipt of customer purchase order.

For any shipments which do not leave the consolidation/distribution center (CDC) within the month following the period in which revenue was recognized, revenue as well as the associated cost, were deferred in that accounting month (refer to AI-HW 002 Revenue Recognition for Shipments to Third Party Owned Consolidation/Distribution Centers).

Bill and Hold transactions

A bill-and-hold arrangement is a contract under which IBM bills a customer for a product but IBM retains physical possession of the product (e.g., the product is shipped to an IBM site or a third party site that is rented, subsidized or otherwise controlled by IBM) until it is transferred to the customer at a point in time in the future. IBM should determine if it has satisfied its performance obligation to transfer a product by evaluating when a customer obtains control of that product by evaluating the control indicators.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Revenue Recognition (SE004.P)

(Continued)

Orders financed via an outside third party require signed Third Party Agreements to be considered firm. Firm order documentation must be present prior to product shipment. Exception approval to ship products without a firm order must be obtained.

Firm order exception (FOE) approval to ship does not alleviate the responsibility to secure firm order documentation by month end. Otherwise, revenue cannot be recognized for that period.

Special attention should be given to:

- Manual revenue billings and accruals recorded at quarter end
- Cancelled contracts and the accounting for termination fees
- Modified installation terms
- Exceptions to standard provisions for ROL (Risk of loss)
- Trade-in (Buy back) of used IBM products from a customer

For OIOs contracts, verify that multiple element arrangements (MEAs) assessment was performed in order to evaluate and record revenue deferrals.

Check that the appropriate channel code was used in order to record revenue and cost in IBM books, specifically for Digital Sales transactions.

IGF considerations

For deals that resulted in a cash transaction, financing proposals and negotiation can resume post quarter close. An IGF financing arranged after quarter close will not lead to IBM revenue recognition issues, provided the lease is a capital lease (for hardware transactions), conforms to INT REV-2, Interpretation of AP REV, Revenue Recognition on Deals with Extended Payment Terms, and does not contain a rate buy down. Leases signed post quarter must close prior to customer payment so the transaction is a new lease rather than a sale and leaseback. Leases closed after the customer has paid the supplier or 90 days from date of invoice are to adhere to sale and leaseback guidelines. IGF Accounting will review IGF financing transactions (related to deals IBM signed prior to quarter close) to confirm that mainline revenue recognition during the quarter was appropriate. Financings signed (in the subsequent quarter) which include either terms that (a) result in an operating lease, (b) contain rate buy down requirements, or (c) result in extended payment terms (INT REV-2) would result in revenue adjustments in the quarter. Refer to AP LESSOR and AP REV-MEA for additional details.

For software, the following additional checks have to be performed:

- Product must be available to the customer. Delivery methods: physical shipment, web availability, keys, provisioning of SaaS and MLC/support and subscription – delivery is over time.
- SW must be available to the customer within the period revenue was recorded. For Software as a Service (SaaS) provisioning of the software must have started.
- If contract includes product substitution clause, all products in the substitution list must be made available to the client within the current period. Substitution lists entitle customers to mix and match undeployed products for other products from the substitution list. Also:
 - no products should be added to list after contract signed.
 - unannounced products cannot be included in the substitution list.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Revenue Recognition (SE004.P)

(Continued)

- For "All You Can Eat" (AYCE) offerings:
 - Subscription and support (S&S) revenue was recognized ratably over the years of the agreement.
 - Licenses: when sold in an ELA, upfront revenue recognition is warranted as long as there is a history of reasonable and reliable deployment estimates and assertions of future outcomes made at the inception of the ELA.

Refer to CRMfS 60-00-00 (SE011.P) for approach details for Accounts Receivable and Billing for Services.

For SW Licenses verify:

- Anniversary reconciliations of the ELA contracts are performed
- Was the SW usage monitored
- For MLC in ELAs (SW specific) in particular, verify that when actual consumption exceeded contractual cap level, the additional SW license fees were actually invoiced (Reference Billing – Ship / Bill / Invoice Reconciliation BI002.P)

Concessions

For accounting purposes, the definition of a concession focuses on whether the concession relates to a good or service committed in the original arrangement and whether it would have impacted the amount of revenue recognized on the contract to date if known at contract inception. Concessions may also be a good indicator for any solution design issues or concerns.

Refer to AI CB-013 - Accounting for Concessions – to validate if concessions were properly accounted.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Secondary Test: Systems Access and Separation of Duties (SE005.S)

Objective:

To ensure that access to IBM systems is controlled in a timely manner, and to prevent unauthorized access to IBM applications and information. Appropriate management approvals should be granted and documented according to business need. To ensure that separation of duties issues are identified and controlled in an appropriate manner to prevent unauthorized transactions from remaining undetected. Secondary controls should be in place where conflicting tasks exist and should be documented in the SOD assessment.

Approach:

Review all SOD assessments, including job roles assigned to employees. Identify secondary controls within the SOD assessment and validate if executed as intended.

Review the list of key applications and databases. Review access lists to establish business need and completeness of CBNs performed. Review the process surrounding access controls for the key applications supporting the audited process. In particular, review access to customer data. Include both IBM and customers systems in scope where appropriate.

Specific items to be aware of:

- Excessive access granted to personal data held on behalf of the customer, and no secondary controls in place to mitigate.
- Failure to obtain all necessary ASCA certifications for a tool used as a trusted source of information.
- Management involvement in prompt access authorization, revalidation, and removal.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Customer Claims/Complaints (SE006.P)

Objective:

Determine if customer correspondence indicates any customer claims or complaints were caused by incorrect marketing practices or verbal commitments.

Approach:

Review and understand the processes for handling and managing customer complaints.

Review customer correspondence for any indication of customer complaints or inappropriate marketing practices.

For the selected contracts / customers, review customer claims and complaints and verify that:

- Claims and complaints were promptly processed.
- Management approval of settlement / resolution exists and is supported by justifying documents.
- Verify that sufficient provisions were taken to account for excessive claims.

Also, determine if any claim or complaint was due to improper or unauthorized commitments to customers, and if actions have been taken to avoid recurrence.

Review documentation to determine if there were indications of recurrent customer issues or business practices violations. Verify that root cause analysis was performed, and appropriate follow-up actions were taken, to minimize the likelihood of similar complaints recurring. Documentation should be maintained of actions taken to research and resolve the complaint.

Data mining in this area can include:

Number of claims, complaints, credit notes, write-offs for a customer, large value credits, excessive delays between original invoice and credit, credits issued in first month following a quarter-end, and credits for duplicate amounts. If reason codes are used for credits issued, select those classified as miscellaneous, unauthorized commitments, product failures, and other nonspecific codes.

Outstanding disputes/claims

Objective: Determine if proper Dispute/Claims Resolution actions assigned to Sales/Sales Management as the Dispute Resolution Owner (DRO) were executed in a timely and effective manner.

Approach: Review DRO actions assigned to Sales/Sales Management to verify that timely and effective resolution activities were executed.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Open Infrastructure Offerings (OIO) (SE007.P)

Objective:

Verify that specific processes and procedures for complex cross-brand deals like open infrastructure offerings (OIO) were followed during the delivery phase. In addition, ensure that all the required reconciliations were performed and approved in a timely manner.

Approach:

For complex cross-brand deals, known as open infrastructure offerings (OIO), specific processes were established, detailing the mandates for sales management and solution delivery. The mandates detailed in the process established at geography level are in general complementary to the usual mandates required for each line of business and performed by a dedicated team. IA teams testing OIO deals have to request copy and be aware of the processes in place and local deviations for these complex, cross-brand deals, regardless the approach of reviewing them for audit purposes (e.g., a specific auditable unit has been defined or you sample OIOs during portfolio CRM audits, or both).

Timing difference between customer payments and revenue recognition. Customer pays according to contractually fixed monthly payment stream. Revenue is recognized when hardware is shipped, or services are performed. A change in delivery schedules without changing the payment schedules will require a recalculation of interest within the OIO and might result in changes to brand revenue allocation.

There is no mandatory identifier in the systems to easily retrieve OIO revenue and cost. Instead, customer and contract numbers are used to identify OIO movements and data must be carefully checked.

For the selected sample of contracts, order letters (OLs) and amendments verify that:

- the contract and all pricing documentation has been provided as required
- conditions of credit checks were fulfilled before delivery start
- existing contracts were transferred into the OIO accurately and in a timely manner
- all contract elements were validated and properly set up in the fulfillment tools
- the payment stream was validated against the signed contract and the pricing model
- OLs and amendments were on file for each contract and were properly reflected in the fulfillment systems
- internal billing and external invoicing occurred in a timely and accurate manner, based on expected schedules
- manual invoicing was properly documented, approved, and delays substantiated
- third-party services and products were delivered according to agreements, and payments were made based on proof of delivery
- customer complaints were reported and processed according to the complaint process
- claims were executed according to the claims process and settlements were appropriately approved
- deal quality reviews were performed appropriately, issues were reported, and actions to fix the issues were followed up in a timely manner.

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: Open Infrastructure Offerings (OIO) (SE007.P)

(Continued)

Signings and Revenue Recognition

For a risk-based sample of contracts, verify whether signings and revenue were recognized in accordance with IBM guidelines and accounting principles. In particular:

- the signing recognition was in line with the WW OIO signing guidelines
- the revenue was recognized in accordance with accounting guidelines (revenue recognition rules, labor claiming, special T&Cs, revenue splits between brands, rebate provisions, etc.)

Verify that supporting documentation for significant manual revenue accruals and check that definitive ledger entry booked through feeder systems (internal or external billing), occurred in the month following the accrual.

Reconciliations

Determine whether the following reconciliations were performed accurately and timely:

- contract reconciliation (reconciliation of movements in fulfilment systems against contract)
- B/S reconciliation
- reconciliation of external invoices vs. contract terms and payment stream, i.e., check that external invoices have been issued based on payment terms, etc.
- validation of B/S account balance through the reconciliation of internal billing and external invoices versus pricing model
- in addition, verify whether required corrections were followed up and performed

CUSTOMER RELATIONSHIP MANAGEMENT FOR HARDWARE AND SOFTWARE AUDIT PROGRAM #30-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Solution Delivery

Primary Test: TEL Services Specific (SE008.P)

Objective:

Ensure that labor claiming is accurate, in accordance with IBM and line of business guidelines, and controls are in place to preserve the integrity of billing and measurements.

Approach:

For TEL Services, refer to the CRMfS (60-00-00) for detailed approach information.

For Labor claiming, refer to CRMfS (60-00-00) - (SE006.P).

Focus on labor claimed prior to contract signature without an authorized pre-contract work (PCW).

Missing or late labor/employee concur expense information items may not be aligned between the BMS and ledger interface. The financial records (costs) will not be booked accordingly. If the items are not cleared in the interface, financial reporting of those related account IDs will be incorrect, resulting in incorrect cost, too high profitability, and incorrect incentives due to employees.

For more accurate information, include the contract number, account ID, and the organization (brand) when submitting the data analysis request to the Technology Team. This will allow better alignment to the database where the records are stored.